

2020/0809 이종민

- 1.103** Calculate the natural frequency and damping ratio for the system in Figure P1.103 given the values $m = 10 \text{ kg}$, $c = 100 \text{ kg/s}$, $k_1 = 4000 \text{ N/m}$, $k_2 = 200 \text{ N/m}$ and $k_3 = 1000 \text{ N/m}$. Assume that no friction acts on the rollers. Is the system overdamped, critically damped or underdamped?

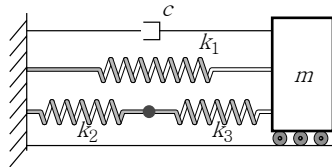


Figure P1.103

$$k_{eq} = \left(\frac{1}{k_2} + \frac{1}{k_3} \right)^{-1} + k_1$$

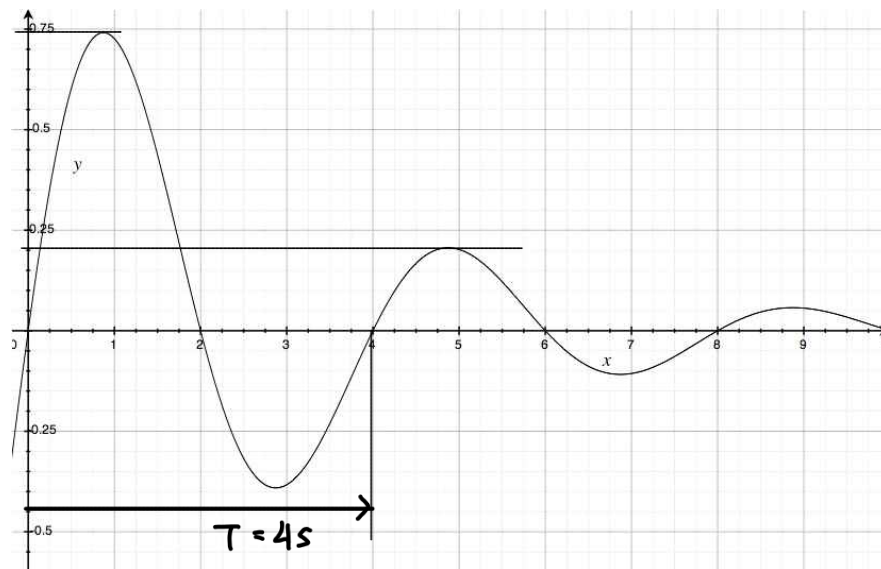
$$= 166.667 + 4000$$

$$= 4166.667 \text{ N/m}$$

$$\therefore \zeta = \frac{c}{c_c} = \frac{100}{2\sqrt{mk_{eq}}} = 0.122 < 1 \text{ (underdamped)}$$

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- 1.106 The displacement of a vibrating spring-mass-damper system is recorded on an $x - y$ plotter and reproduced in Figure P1.106. The y coordinate is the displacement in cm and the x coordinate is time in seconds. From the plot determine the natural frequency, the damping ratio and the damped natural frequency. $\omega_n, \zeta, \omega_d$



P1.106 A plot of displacement versus time for a vibrating system.

첫번째 점선면 때 시작하면

$$x(t) = 0.75 \text{ cm}$$

$$x(t+T) = 0.2 \text{ cm}$$

mass, spring, damper system의 $x(t)$ 는 다음과 같다.

$$x(t) = A e^{-\zeta \omega_n t} \sin(\omega_d t + \phi)$$

$$\therefore \zeta = \ln \frac{x(t)}{x(t+T)} = \ln \left(\frac{A e^{-\zeta \omega_n t} \sin(\omega_d t + \phi)}{A e^{-\zeta \omega_n (t+T)} \sin(\omega_d (t+T) + \phi)} \right)$$

$$\ln \frac{0.75}{0.2} = \ln e^{\zeta \omega_n T}$$

$$1.3217 = \zeta \omega_n T \quad \leftarrow T = \frac{2\pi}{\omega_d} = \frac{2\pi}{\omega_n \sqrt{1-\zeta^2}}$$

$$1.3217 = \frac{2\pi \zeta}{\sqrt{1-\zeta^2}}$$

$$\therefore \zeta = 0.344 \quad \therefore \omega_n = 0.947 \text{ rad/s}, \omega_d = 0.887 \text{ rad/s}$$

2020/0809 아중

1.107 Show that the logarithmic decrement is equal to

$$\delta = \frac{1}{n} \ln \frac{x_0}{x_n}$$

where x_n is the amplitude of vibration after n cycles have elapsed.

$$\delta = \ln \frac{x_0}{x_1}$$

$$\delta = \ln \frac{x_1}{x_2}$$

$\vdots$

$$\delta = \ln \frac{x_{n-1}}{x_n}$$

$$\therefore \sum_{i=1}^n \delta_i = \sum_{i=1}^n \ln \frac{x_{i-1}}{x_i}$$

$$n\delta = \ln \frac{x_0}{x_1} + \ln \frac{x_1}{x_2} + \dots + \ln \frac{x_{n-1}}{x_n}$$

$$= \ln \left(\frac{x_0}{x_1} \times \frac{x_1}{x_2} \times \dots \times \frac{x_{n-1}}{x_n} \right)$$

$$n\delta = \ln \frac{x_0}{x_n}$$

$$\delta = \frac{1}{n} \ln \frac{x_0}{x_n}$$

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- 1.123** Consider the system of Figure P1.123. (a) Write the equations of motion in terms of the angle, θ , the bar makes with the vertical. Assume linear deflections of the springs and linearize the equations of motion. (b) Discuss the stability of the linear system's solutions in terms of the physical constants, m , k , and ℓ . Assume the mass of the rod acts at the center as indicated in the figure.

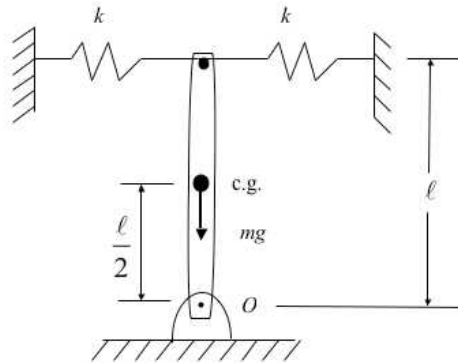
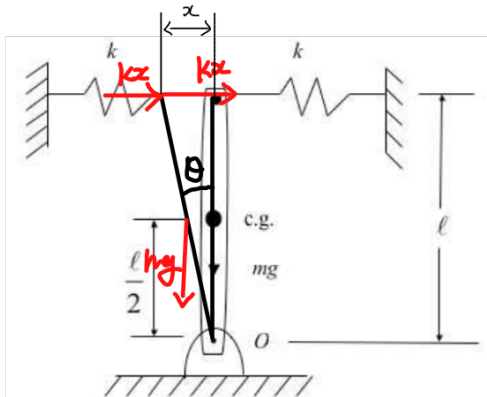


Figure P1.123



$$\Sigma M_o = J\ddot{\theta}$$

$$J\ddot{\theta} = \frac{l}{2} \sin\theta \times mg - l \times 2kx \quad \leftarrow x = l\sin\theta$$

$$J\ddot{\theta} = \frac{lmg}{2} \sin\theta - 2l^2k \sin\theta$$

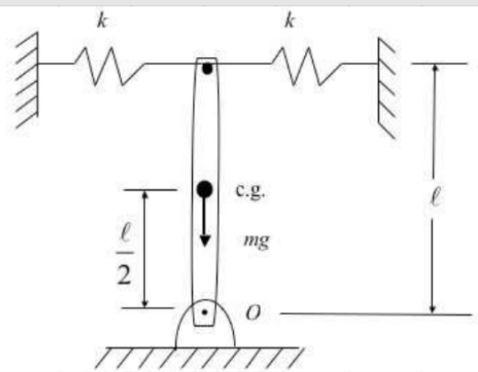
$$\theta \approx 0 \text{ 일때 } \sin\theta \approx \theta$$

$$J\ddot{\theta} = \left(\frac{lmg}{2} - 2l^2k \right) \theta$$

$$J = m\left(\frac{l}{2}\right)^2$$

$$\therefore \frac{ml^2}{4} \ddot{\theta} + \left(2l^2k - \frac{lmg}{2} \right) \theta = 0$$

$$\ddot{\theta} + \left(\frac{8k}{m} - \frac{2g}{l} \right) \theta = 0$$



$$T_{\text{rot}} = \frac{1}{2} I \dot{\theta}^2$$

$$= \frac{m l^2}{8} \dot{\theta}^2$$

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$$U_p = \frac{1}{2} k x^2 \times 2$$

$$= k l^2 \theta^2$$

$$U_h = \frac{1}{2} m g h = \frac{m g}{2} \left(1 - \frac{\theta^2}{2}\right) = \frac{m g}{2} - \frac{m g}{4} \theta^2$$

$$L = T - U = \frac{m l^2}{8} \dot{\theta}^2 + \frac{m g}{4} \theta^2 - k l^2 \theta^2 - \frac{m g}{2}$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = 0$$

$$\frac{m l^2}{4} \ddot{\theta} + \left(2 k l^2 - \frac{m g}{2} \right) \theta = 0$$

$$\ddot{\theta} + \left(\frac{8 k}{m} - \frac{2 g}{l} \right) \theta = 0$$
